Beyond Microprice

Martingale Correctors and Value Functions from the Limit Order Book

Lev Petersen

Abstract. Stoikov (2018) introduced the *microprice* as a horizon-free limit of conditional expectations of future prices computed from Level-I order-book data via an absorbing Markov chain. We take that construction as a starting point and push beyond it: at mid-price move times the problem becomes a finite Markov additive system, and the microprice correction is one instance of a broader Poisson-corrector framework.
The key object is the one-move conditional drift G_1 of the mid-price on the embedded post-move chain B. When the stationary drift vanishes, Stoikov's adjustment is exactly the unique centered solution of the Poisson equation $(I - B)g = G_1$. Consequently $p_k + g(X_k)$ is a martingale. The same Poisson equation yields martingale correctors for arbitrary additive order-book functionals.
We give transparent conditions for the absorbing fundamental matrix $(I - Q)^{-1}$ and the post-move chain B, prove convergence and truncation bounds for the microprice series, derive an exact finite-horizon decomposition that separates stationary drift from the centered corrector, and record a mean-square optimality identity against state-based baselines. We also introduce discounted resolvent regularization, which is always well posed, and quantify its bias as the discount tends to one. Empirically, we first reproduce Stoikov's BAC/CVX March 2011 public-data benchmark, then test the broader family on five names from the raw WSELOB-2017 limit-order-book data. On that out-of-sample panel the weighted mid is uniformly dominated, the price corrector remains close to a move-time martingale, and the discounted resolvent is again nearly indistinguishable from the centered Poisson correction.

Keywords: microprice, martingale corrector, Poisson equation, Markov additive process, Markov potential, value function, Neumann series, resolvent, absorbing Markov chain

Contents

1 Introduction **4**

2 Setup and notation **4**

2.1 Prices and book features 4

2.2 Event time vs. move time 5

2.3 State discretization 5

2.4 Standing assumptions 5

3 Microprice mechanics: from absorbing event time to a Poisson equation **6**

3.1 Transition decomposition 6

3.2 The fundamental matrix of the no-move kernel 7

3.3 Post-move chain and first adjustment 8

3.4 The microprice adjustment as a Poisson equation 8

3.5 Microprice as a martingale corrector 10

3.6 Prediction optimality and mean-square improvement 11

3.7 A regularized microprice via discounted resolvents 12

4 Microprice as a general martingale constructor **13**

4.1 Doob decomposition for state observables 13

4.2 Poisson correctors for additive functionals 13

4.3 Operator notation: potentials, projections, and generalized "micro-" transforms . . . 14

4.4 Vector-valued functionals 15

5 Empirical highlights beyond the baseline **15**

5.1 Reproducing Stoikov on the public data 15

5.2 What is actually new 16

5.3 Venue-level stress check: public IEX TOPS 17

6 Discussion and limitations **18**

7 Conclusion **18**

A Proofs **20**

1 Introduction

Stoikov's microprice [2] starts from a practical microstructure question: given the current state of the limit order book, what is a fair estimate of the asset's value? The classical mid-price

$$p_t := \frac{b_t + a_t}{2}$$

ignores information in queue imbalance and related order-flow imbalance signals [4, 3]. A fair value should incorporate how the current book state predicts future mid-price moves.

This paper isolates the operator structure behind that prediction problem. At *move times* (times when the mid-price changes), the pair consisting of the book state and the price increment forms a finite Markov additive system. The one-step predictable drift of the price is state dependent. The microprice correction is precisely the Poisson corrector that removes this drift.

Three consequences are central. First, the absorbing-chain computation used by Stoikov can be written cleanly in terms of the no-move kernel Q, the post-move chain B, and the one-move drift G_1. Second, when the stationary drift per move vanishes, the infinite-horizon correction is the unique centered solution of $(I - B)g = G_1$, so price plus corrector is a martingale. Third, even when the stationary drift does not vanish, the same Poisson equation separates the linear trend from a bounded state correction; this gives the correct martingale decomposition and explains exactly why the undiscounted microprice diverges in trending regimes. The broader message is that the microprice is just the first example in a larger operator family.

Roadmap. Section 2 fixes the two clocks (event time and move time) and the Markov assumptions. Section 3 derives G_1 and B from the absorbing event-time dynamics, proves the Poisson-equation representation, and records finite-horizon, prediction, and discounted-resolvent consequences. Section 4 abstracts the same argument to general additive functionals. The companion note [1] isolates the discounted value-function viewpoint.

2 Setup and notation

We work with Level-I order book data. At each event time t we observe the best bid b_t, best ask a_t, and their queue sizes B_t and A_t.

2.1 Prices and book features

We use the following derived quantities:

$$\begin{aligned}
\text{Mid-price:} \quad & p_t = (b_t + a_t)/2, \\
\text{Spread:} \quad & s_t = a_t - b_t, \\
\text{Imbalance:} \quad & i_t = \frac{B_t - A_t}{B_t + A_t} \in [-1, 1], \\
\text{Size-weighted mid:} \quad & m_t = \frac{a_t B_t + b_t A_t}{B_t + A_t} = p_t + i_t \cdot \frac{s_t}{2}.
\end{aligned} \tag{1}$$

Mapping to Stoikov (2018) notation. Stoikov [2] uses an imbalance $I_S \in [0,1]$ and the *half-spread* $S = \frac{1}{2}(a_t - b_t)$. Our imbalance $i_t \in [-1,1]$ and *full spread* $s_t = a_t - b_t$ are related by

$$i_t = 2I_S - 1, \qquad s_t = 2S.$$

With this identification, the size-weighted mid identity agrees exactly with Stoikov's weighted mid:

$$m_t = p_t + i_t \frac{s_t}{2} = p_t + (2I_S - 1)S.$$

The object of interest is not m_t itself, but rather a *fair value* obtained by conditioning on the book state and looking forward to future price moves.

2.2 Event time vs. move time

There are two natural clocks. Event time counts every book update (queue change, trade, etc.); the discretized state evolves as a Markov chain at this clock. Move time counts only events that change the mid-price, and this embedded chain is the right object for the microprice/martingale-corrector viewpoint.

Let $\tau_0 = 0$ and let

$$\tau_{k+1} := \inf\{t > \tau_k : p_t \neq p_{t-}\} \tag{2}$$

be the $(k+1)$-st mid-price move time. Define the move-time price and state

$$p_k := p_{\tau_k}, \qquad X_k := X_{\tau_k}.$$

We write $\mathcal{F}_k := \sigma\big((p_j, X_j)_{0 \le j \le k}\big)$ for the *move-time filtration*. All "fair value" objects in this paper are conditional expectations given the information available at time k, i.e. conditional on $\mathcal{F}_k$ (not merely on X_k).

Because the mid-price is additive across moves, it is natural to forecast *increments* relative to the current price level. Under the Markov-additive structure formalized in Assumption 3, the conditional law of future price increments depends on the book state X_k but not on the absolute price level p_k. Consequently, many conditional expectations take the form "current value + state-dependent correction".

2.3 State discretization

We discretize continuous features into buckets. For the single-instrument setting, a common choice is

$$X_t = (\text{spread bucket}, \text{imbalance bucket}),$$

with m spread buckets and n imbalance buckets, giving $L = mn$ states. Nothing in the operator algebra depends on this particular choice; any finite Markov state can be used.

2.4 Standing assumptions

Assumption 1 (Markov property)**.** The discretized state X_t is Markov at event times.

Assumption 2 (Bounded increments)**.** One-step mid-price changes satisfy $|\Delta p_t| \le D$ for some constant D.

Assumption 3 (Markov additive structure / translation invariance)**.** At move times, the pair (X_k, p_k) is a *Markov additive process* in the following sense: conditional on $\mathcal{F}_k$, the law of $(X_{k+1}, \Delta p_{k+1})$ depends on the past only through X_k, and the conditional distribution of the increment $\Delta p_{k+1} := p_{k+1} - p_k$ depends on X_k but not on the absolute price level p_k. In particular, the conditional mean increment is state dependent: there exists a deterministic function $m : \mathcal{X} \to \mathbb{R}$ such that

$$\mathbb{E}[\Delta p_{k+1} \mid \mathcal{F}_k] = m(X_k).$$

In Section 3 this function is identified with the first adjustment G_1 produced by the absorbing-chain construction.

Assumption 4 (Ergodicity at move times)**.** The post-move chain ($B := (I - Q)^{-1}T$, defined in Section 3) is irreducible and aperiodic, admitting a unique stationary distribution.

3 Microprice mechanics: from absorbing event time to a Poisson equation

This section makes explicit the mechanics behind Stoikov's microprice computation [2]. We first decompose the event-time kernel into no-move and move pieces (Q, T), then use the fundamental matrix $(I-Q)^{-1}$ to compute the embedded post-move chain B and the one-move drift G_1, and finally recognize the infinite-horizon adjustment as the centered solution to a Poisson equation on B.

3.1 Transition decomposition

Partition event-time transitions according to whether the mid-price moves. Let Q_{xy} denote the probability of transitioning from state x to y without a price move, and T_{xy} the probability with a move. The full event-time transition matrix is

$$P := Q + T,$$

and is row-stochastic. Whenever stopping-time arguments are needed, we write

$$\mathcal{E}_n := \sigma\big((X_j, p_j)_{0 \le j \le n}\big)$$

for the event-time filtration.

Let

$$p_{\text{move}}(x) := \sum_y T_{xy} = 1 - \sum_y Q_{xy}$$

be the probability the next event is a mid-price move.

For price changes, we allow the (conditional) move size to depend on the *move transition*. Let

$$\Delta^p_{xy} := \mathbb{E}\big[\Delta p \,\big|\, X_{n-1} = x,\ X_n = y,\ \text{the step } (x \to y) \text{ is a price move}\big]$$

denote the conditional mean mid-price change associated with a move transition $x \to y$. Collect these means into the matrix $\Delta^p \in \mathbb{R}^{L \times L}$. Then the unconditional one-step drift at event time (including zero change on no-move events) is

$$\bar{r}_x := \mathbb{E}[\Delta p_n \mid X_{n-1} = x] = \sum_y T_{xy}\, \Delta^p_{xy}, \qquad \text{i.e.} \qquad \bar{r} = (T \odot \Delta^p)\,\mathbf{1}, \tag{3}$$

where $\odot$ denotes entrywise (Hadamard) product and $\mathbf{1}$ is the all-ones column vector.

A common special case is that the move size distribution depends only on the *pre-move* state x and takes values $\{\delta_j\}_j$ (e.g. $\pm$ one tick). Writing $R_{xj} := \mathbb{P}(\Delta p = \delta_j \mid X_{n-1} = x,\ \text{move})$ and $\boldsymbol{\delta}$ for the vector with entries δ_j, the conditional mean move size is

$$r_x^{\text{move}} := \sum_j R_{xj}\,\delta_j, \qquad \text{i.e.} \qquad r^{\text{move}} = R\,\boldsymbol{\delta}.$$

In this factorized setting $\Delta_{xy}^p = r_x^{\text{move}}$ does not depend on y and (3) reduces to

$$\bar{r}_x = p_{\text{move}}(x)\, r_x^{\text{move}}, \qquad \text{i.e.} \qquad \bar{r} = D_{p_{\text{move}}}\, r^{\text{move}},$$

where $D_{p_{\text{move}}} := \operatorname{diag}(p_{\text{move}}(x))$.

3.2 The fundamental matrix of the no-move kernel

Between consecutive price moves, the book state evolves according to the no-move kernel Q. Understanding this transient dynamics is essential: the fundamental matrix $(I - Q)^{-1}$ aggregates expected visits before the next move, and its invertibility guarantees that moves eventually arrive from every state.

The no-move matrix Q is substochastic: each row sums to at most 1. When moves occur with uniformly positive probability, the spectral radius of Q is strictly less than one, and the fundamental matrix exists.

Definition 3.1. A matrix $Q \in \mathbb{R}^{L\times L}$ is *substochastic* if $Q_{xy} \geq 0$ for all x, y and $\sum_y Q_{xy} \leq 1$ for all x.

Lemma 3.2 (Spectral radius bound)**.** *Let Q be substochastic with $\sum_y Q_{xy} \leq 1 - \varepsilon$ for all x. Then $\rho(Q) \leq 1 - \varepsilon < 1$.*

Proposition 3.3 (Fundamental matrix)**.** *Under the conditions of Lemma 3.2, the matrix $I - Q$ is invertible with*

$$N := (I - Q)^{-1} = \sum_{k=0}^{\infty} Q^k, \qquad \|N\|_\infty \leq \frac{1}{\varepsilon}.$$

The entries N_{xy} equal the expected number of visits to y, starting from x, before the first price move [5].

Remark 3.4 (Sufficient versus necessary conditions for $I - Q$ invertibility). Lemma 3.2 assumes a uniform bound $\sum_y Q_{xy} \leq 1 - \varepsilon$, i.e. a state-independent lower bound on move probability. This is a convenient *sufficient* condition yielding the clean norm bound $\|N\|_\infty \leq 1/\varepsilon$. For the microprice algebra itself, the essential requirement is the *transience* of the no-move subchain, equivalently $\rho(Q) < 1$ (eventual arrival of a move almost surely from every state).

Lemma 3.5 (Non-negativity)**.** *The fundamental matrix N has non-negative entries, with $N_{xx} \geq 1$ for all x.*

3.3 Post-move chain and first adjustment

Definition 3.6. The *first adjustment* is the expected cumulative price change up to the first move:

$$G_1 := (I - Q)^{-1}\bar{r} = N\,\bar{r}.$$

In the factorized setting $\bar{r} = D_{p_{\text{move}}} r^{\text{move}}$ (see (3)), this reads $G_1 = N D_{p_{\text{move}}} r^{\text{move}}$. The *post-move chain* is the transition matrix observed only at move times:

$$B := (I - Q)^{-1} T = N\,T.$$

Lemma 3.7 (Embedded-chain interpretation)**.** *Let $(X_n)_{n\geq 0}$ be the event-time Markov chain with transition matrix $P = Q + T$, and let*

$$\tau := \inf\{n \geq 1 : \textit{the step } (X_{n-1} \to X_n) \textit{ is a price move}\}.$$

Then $B_{xy} = \mathbb{P}_x(X_\tau = y)$ is the distribution of the post-move state, and the first adjustment is

$$G_1(x) = \mathbb{E}_x[\Delta p_\tau],$$

the expected mid-price change at the next move. Equivalently, G_1 is the unique solution of the first-step system

$$G_1 = \bar{r} + QG_1. \tag{4}$$

Lemma 3.8 (Move-time chain is Markov with kernel B)**.** *Let $(X_n)_{n\geq 0}$ be the event-time Markov chain with transition matrix $P = Q + T$, and let $(\tau_k)_{k\geq 0}$ be the successive move times as in (2). Then the sampled process $(X_{\tau_k})_{k\geq 0}$ is a Markov chain on $\mathcal{X}$ with transition kernel B:*

$$\mathbb{P}\big(X_{\tau_{k+1}} = y \mid \mathcal{E}_{\tau_k}\big) = B_{X_{\tau_k},y} \qquad a.s.$$

In particular, conditional on the full event-time history up to τ_k, the law of the next post-move state depends only on X_{τ_k}.

Lemma 3.9. *If $Q + T$ is row-stochastic and $\rho(Q) < 1$, then B is row-stochastic.*

3.4 The microprice adjustment as a Poisson equation

Fix a move-time index k and recall the move-time filtration $\mathcal{F}_k = \sigma\big((p_j, X_j)_{0\leq j\leq k}\big)$. For $n \geq 0$, define the *n-move fair value* (the best mean-square predictor of the future mid-price at horizon n moves) by

$$p_k^{(n)} := \mathbb{E}\big[p_{k+n} \mid \mathcal{F}_k\big].$$

Because p_k is $\mathcal{F}_k$-measurable and the mid-price is additive across moves, we can write

$$p_{k+n} = p_k + \sum_{\ell=1}^{n} \Delta p_{k+\ell}, \qquad \Delta p_{k+\ell} := p_{k+\ell} - p_{k+\ell-1}.$$

Under the Markov-additive/translation-invariance Assumption 3, the conditional law of future increments given $\mathcal{F}_k$ depends on the past only through X_k. Hence there exists a

deterministic function $G^{(n)} : \mathcal{X} \to \mathbb{R}$ such that

$$p_k^{(n)} = p_k + G^{(n)}(X_k), \qquad \text{where} \qquad G^{(n)}(x) := \mathbb{E}\left[p_{k+n} - p_k \,\middle|\, X_k = x\right]. \tag{5}$$

By time-homogeneity of the move-time dynamics, $G^{(n)}$ does not depend on k.

Remark 3.10 (Conditioning on X_k versus conditioning on $\mathcal{F}_k$). If one conditions only on X_k (and not on the current price level), then $\mathbb{E}[p_{k+n} \mid X_k] = \mathbb{E}[p_k \mid X_k] + G^{(n)}(X_k)$. In trading applications, p_k is observed, so the relevant fair value is (5).

When the limit exists, the *microprice* (horizon-free fair value at move times) is

$$p_k^{\text{micro}} := \lim_{n\to\infty} p_k^{(n)} = p_k + G^*(X_k), \qquad \text{where} \qquad G^* := \lim_{n\to\infty} G^{(n)}.$$

By the tower property and the Markov property at move times, the adjustments satisfy the recursion

$$G^{(0)} = 0, \qquad G^{(n)} = G_1 + BG^{(n-1)}.$$

Unrolling gives

$$G^{(n)} = \sum_{\ell=0}^{n-1} B^\ell G_1.$$

When it converges, the infinite-horizon adjustment is

$$G^* := \lim_{n\to\infty} G^{(n)} = \sum_{\ell=0}^{\infty} B^\ell G_1. \tag{6}$$

Since B is stochastic (hence has eigenvalue 1), the convergence of (6) requires a *centering* condition.

Definition 3.11 (Centering)**.** Let π be the stationary distribution of B. We say G_1 is *centered* if $\pi^T G_1 = 0$. Equivalently, if the move-time chain is started from stationarity, then the mean price increment per move vanishes:

$$\mathbb{E}_\pi[\Delta p_{k+1}] = \pi^T G_1 = 0.$$

Remark 3.12 (Equivalence with Stoikov's $B^* G_1 = 0$ condition). For a finite ergodic chain, $B^k \to \mathbf{1}\pi^T$ as $k \to \infty$, so Stoikov's limit kernel satisfies $B^* = \mathbf{1}\pi^T$. Hence $B^* G_1 = 0$ is equivalent to $\pi^T G_1 = 0$.

Theorem 3.13 (Microprice convergence)**.** *Let B be an ergodic Markov kernel on a finite state space with stationary distribution π. Let*

$$\lambda_* := \max\{|\lambda| : \lambda \in \sigma(B),\ \lambda \neq 1\} \in [0, 1)$$

be the second-largest eigenvalue modulus (SLEM). See, e.g., [7, 8] for background on the spectral picture of finite ergodic chains. If G_1 is centered, i.e. $\pi^T G_1 = 0$, then:

1. *The series (6) converges absolutely to*

$$G^* = (I - B + \mathbf{1}\pi^T)^{-1} G_1;$$

see [5, 6]. In particular, G^* is centered: $\pi^T G^* = 0$.

2. *G^* is the unique solution of the centered Poisson equation*

$$(I - B)G^* = G_1, \qquad \pi^T G^* = 0. \tag{7}$$

3. *For any $\rho \in (\lambda_*, 1)$ there exists $C_\rho < \infty$ such that the truncation error after n terms satisfies*

$$\left\| G^* - \sum_{k=0}^{n-1} B^k G_1 \right\|_\infty \le \frac{C_\rho \rho^n}{1-\rho} \|G_1\|_\infty.$$

Proposition 3.14 (Finite-horizon decomposition and trend term)**.** *Let $\mu := \pi^T G_1$ and write $h := G_1 - \mu \mathbf{1}$. Assume B is ergodic, and let $\tilde{G}$ be the unique centered solution of*

$$(I - B)\tilde{G} = h, \qquad \pi^T \tilde{G} = 0.$$

Then for every $n \ge 1$,

$$G^{(n)} = n\mu \mathbf{1} + \tilde{G} - B^n \tilde{G}. \tag{8}$$

In particular, in the centered regime $\mu = 0$ and $\tilde{G} = G^*$, so

$$G^{(n)} = G^* - B^n G^*.$$

3.5 Microprice as a martingale corrector

At move times, (X_k, p_k) is a Markov additive process:

$$\mathbb{P}(X_{k+1} = y \mid X_k = x) = B_{xy}, \qquad \mathbb{E}[\Delta p_{k+1} \mid X_k = x] = G_1(x),$$

where $\Delta p_{k+1} := p_{k+1} - p_k$.

Theorem 3.15 (Microprice martingale)**.** *Assume the centered Poisson equation (7) has solution G^* (equivalently, $\pi^T G_1 = 0$). Then the move-time process*

$$\mathcal{P}_k := p_k^{\text{micro}} := p_k + G^*(X_k) \tag{9}$$

is a martingale with respect to the move-time filtration.

Remark 3.16 (Trending markets and discounted surrogates). When $\mu := \pi^T G_1 \neq 0$, Proposition 3.14 shows that the fair-value adjustment grows as $n\mu \mathbf{1}$ plus a bounded state term. Thus the undiscounted microprice is not finite, but the centered corrector $\tilde{G}$ still yields the martingale decomposition

$$p_k - k\mu + \tilde{G}(X_k)$$

up to an additive constant. Discounted resolvents therefore play a regularizing role in nonstationary or weakly centered regimes.

Proposition 3.17 (First adjustment bounds)**.** *If the mid-price move size is uniformly bounded by $|\Delta p| \le D$, then $\|G_1\|_\infty \le D$.*

If one instead bounds via the operator norm, $\|G_1\|_\infty \le \|N\|_\infty \|\bar{r}\|_\infty$, and imposes an absorption floor $p_{\text{move}}(x) \ge \varepsilon$ so that $\|N\|_\infty \le 1/\varepsilon$, then $\|G_1\|_\infty \le \|\bar{r}\|_\infty/\varepsilon$.

3.6 Prediction optimality and mean-square improvement

For a fixed horizon $n \ge 1$, the n-move fair value $p_k^{(n)} = \mathbb{E}[p_{k+n} \mid \mathcal{F}_k]$ is, by definition, the conditional expectation of the future mid-price given the information available at move time k. This elementary fact already yields a strong *dominance* statement: $p_k^{(n)}$ minimizes mean-squared prediction error among *all* $\mathcal{F}_k$-measurable predictors. Specializing to predictors of the form "current mid-price + state-based correction" yields a clean improvement guarantee over baselines such as the mid-price itself or the size-weighted mid.

Proposition 3.18 (Pythagorean identity for conditional expectations)**.** *Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra and let $Z \in L^2$. Write $\hat{Z} := \mathbb{E}[Z \mid \mathcal{G}]$. Then for any $\mathcal{G}$-measurable $Y \in L^2$,*

$$\mathbb{E}\big[(Z - Y)^2\big] = \mathbb{E}\big[(Z - \hat{Z})^2\big] + \mathbb{E}\big[(\hat{Z} - Y)^2\big]. \tag{10}$$

Corollary 3.19 (Dominance over state-based baselines)**.** *Fix $n \ge 1$ and let $\hat{p}_k := p_k + a(X_k)$ for some deterministic function $a : \mathcal{X} \to \mathbb{R}$. Then*

$$\mathbb{E}\big[(p_{k+n} - \hat{p}_k)^2\big] = \mathbb{E}\big[(p_{k+n} - p_k^{(n)})^2\big] + \mathbb{E}\big[(G^{(n)}(X_k) - a(X_k))^2\big], \tag{11}$$

where $G^{(n)}$ is the adjustment from (5). In particular:

- *The mean-square optimal predictor among all $\hat{p}_k = p_k + a(X_k)$ is obtained by taking $a = G^{(n)}$. It is unique up to modification on states with $\mathbb{P}(X_k = x) = 0$; in particular it is unique whenever X_k has full support.*
- *Taking the baseline $a \equiv 0$ (mid-price predictor) gives the strict improvement identity*

  $$\mathbb{E}\big[(p_{k+n} - p_k)^2\big] = \mathbb{E}\big[(p_{k+n} - p_k^{(n)})^2\big] + \mathbb{E}\big[(G^{(n)}(X_k))^2\big],$$

  so the n-move fair value improves MSE over the mid-price by $\mathbb{E}[(G^{(n)}(X_k))^2]$ (unless $G^{(n)} \equiv 0$).
- *If the size-weighted mid m_k is included in the state (so that $m_k = p_k + a_{\text{wm}}(X_k)$ for some a_{wm}), then (11) applies verbatim and $p_k^{(n)}$ dominates m_k in mean-square error.*

Remark 3.20 (As $n \to \infty$ in the centered regime). When the centering condition $\pi^T G_1 = 0$ holds, $G^{(n)} \to G^*$ and $p_k^{(n)} \to p_k^{\text{micro}}$. Equation (11) shows that the *MSE gap* between any fixed state-based baseline $a(X_k)$ and the optimal n-move predictor is exactly $\mathbb{E}[(G^{(n)}(X_k) - a(X_k))^2]$, which converges to $\mathbb{E}[(G^*(X_k) - a(X_k))^2]$ as $n \to \infty$. Thus, in stationary centered regimes the advantage of using the microprice correction saturates to a finite limit even though the absolute prediction error typically grows with horizon.

3.7 A regularized microprice via discounted resolvents

The microprice $G^* = \sum_{k\geq 0} B^k G_1$ is an *undiscounted* potential and therefore requires centering because B has eigenvalue 1. A simple and useful regularization is to introduce a discount factor $\alpha \in (0,1)$, which yields a well-posed resolvent on all of $\mathbb{R}^L$ without any centering.

Definition 3.21 (Discounted adjustment)**.** For $\alpha \in (0,1)$ define the discounted potential

$$G^{(\alpha)} := \sum_{k=0}^{\infty} \alpha^k B^k G_1 = (I - \alpha B)^{-1} G_1, \tag{12}$$

and the associated discounted microprice process

$$p_k^{(\alpha)} := p_k + G^{(\alpha)}(X_k).$$

Proposition 3.22 (Interpretations of the discounted adjustment)**.** *For $\alpha \in (0,1)$ and any move-time index k,*

$$G^{(\alpha)}(X_k) = \mathbb{E}\left[\sum_{j=1}^{\infty} \alpha^{j-1} \Delta p_{k+j} \,\middle|\, \mathcal{F}_k\right]. \tag{13}$$

Moreover, if N is an independent geometric random variable on $\{0,1,2,\dots\}$ with $\mathbb{P}(N = n) = (1-\alpha)\alpha^n$, then

$$\mathbb{E}[p_{k+N} \mid \mathcal{F}_k] = p_k + \alpha(I - \alpha B)^{-1} G_1(X_k) = p_k + \alpha\, G^{(\alpha)}(X_k). \tag{14}$$

Proposition 3.23 (Discount-to-undiscounted limit with quantitative bias control)**.** *Assume B is ergodic and G_1 is centered ($\pi^T G_1 = 0$), so that the undiscounted microprice adjustment G^* exists. Let $\rho \in (\lambda_*, 1)$ and let C_ρ be such that $\|B^k G_1\|_\infty \leq C_\rho \rho^k \|G_1\|_\infty$ for all $k \geq 0$ (as in the proof of Theorem 3.13). Then for every $\alpha \in (0,1)$,*

$$\|G^{(\alpha)} - G^*\|_\infty \leq (1-\alpha)\, \frac{C_\rho\, \rho}{(1-\rho)^2}\, \|G_1\|_\infty.$$

In particular, $G^{(\alpha)} \to G^*$ as $\alpha \uparrow 1$.

Remark 3.24 (Why discounting is useful even when centering holds). In applications B and G_1 are estimated from data and may violate exact centering due to sampling error or nonstationarity. The discounted adjustment $G^{(\alpha)}$ is always well-defined and acts as a stable surrogate: it suppresses long-horizon amplification by the nearly-invariant constant mode and yields explicit control of the bias as $\alpha \uparrow 1$. A systematic discounted value-function treatment is developed in the companion note [1].

4 Microprice as a general martingale constructor

The microprice construction in Section 3 used only two properties of the mid-price: additivity across moves and a state-dependent conditional mean increment. These two properties suffice for a general martingale construction. Any additive functional driven by a Markov state admits a predictable drift term plus a martingale term, and the martingale term is obtained by solving a Poisson equation.

4.1 Doob decomposition for state observables

Let $(X_k)_{k\geq 0}$ be a Markov chain with transition matrix B. For any bounded function $f : \mathcal{X} \to \mathbb{R}$, the process $f(X_k)$ admits the decomposition

$$f(X_k) = f(X_0) + M_k + A_k, \tag{15}$$

where

$$A_{k+1} - A_k = (Bf - f)(X_k), \qquad M_{k+1} - M_k = f(X_{k+1}) - Bf(X_k),$$

and (M_k) is a martingale. Equation (15) is the discrete-time Doob decomposition; see, e.g., [11].

4.2 Poisson correctors for additive functionals

Now let $(Y_k)_{k\geq 0}$ be an *additive* functional driven by the chain:

$$Y_{k+1} = Y_k + \xi_{k+1},$$

where each increment ξ_{k+1} is integrable and

$$\bar{\xi}(x) := \mathbb{E}[\xi_{k+1} \mid X_k = x].$$

Let $\mathcal{G}_k := \sigma\big((X_j, Y_j)_{0\leq j\leq k}\big)$ be the natural filtration of the pair. The stationary mean drift is $\mu := \pi^T \bar{\xi}$. The centered drift is $h := \bar{\xi} - \mu\mathbf{1}$.

Theorem 4.1 (Markov additive martingale corrector)**.** *Assume B is ergodic with stationary distribution π, and define $h := \bar{\xi} - \mu\mathbf{1}$ with $\mu = \pi^T \bar{\xi}$. Let g be the unique centered solution of the Poisson equation*

$$(I - B)g = h, \qquad \pi^T g = 0. \tag{16}$$

Then the compensated process

$$\mathcal{M}_k := Y_k - k\mu + g(X_k) - g(X_0) \tag{17}$$

is a martingale with respect to $(\mathcal{G}_k)_{k\geq 0}$.

Theorem 4.1 is a standard way to construct martingales from additive functionals via Poisson's equation; see, e.g., [9, 10] for general treatments.

The microprice martingale (Theorem 3.15) is the special case $Y_k = p_k$ and $h = G_1 - \mu\mathbf{1}$.

Scope. Theorem 4.1 is purely structural: once an observable is written as an additive functional with conditional mean drift $\bar{\xi}(x)$, solving the Poisson equation (16) produces the drift-corrected martingale. We omit application-specific choices of ξ and focus on the operator construction.

4.3 Operator notation: potentials, projections, and generalized "micro-" transforms

Recasting the Poisson equation in operator language clarifies what the construction really does: the map from drift profiles to correctors is a single linear operator, and all microprice-type constructions are instances of applying it.

It is convenient to phrase the preceding results as statements about linear operators on functions on the state space. Identify a function $f : \mathcal{X} \to \mathbb{R}$ with its coordinate vector in $\mathbb{R}^L$. The post-move chain B then acts as a Markov operator

$$(\mathsf{B}f)(x) := \mathbb{E}[f(X_{k+1}) \mid X_k = x] = \sum_y B_{xy} f(y).$$

Let π be the stationary distribution of B and define the rank-one projection onto constants

$$(\Pi f)(x) := (\pi^T f)\,\mathbf{1}(x), \qquad \text{equivalently} \qquad \Pi = \mathbf{1}\pi^T,$$

so that $\mathrm{Ran}(\Pi) = \mathrm{span}\{\mathbf{1}\}$ and $\ker(\Pi) = \{f : \pi^T f = 0\}$. On the centered subspace $\ker(\Pi)$, the operator $I - \mathsf{B}$ is invertible and its inverse is the *potential* (or *Poisson-solver*) operator

$$\mathsf{U} := (I - \mathsf{B})^{-1} \text{ on } \ker(\Pi), \qquad \text{so that} \qquad \mathsf{U}h = \sum_{n=0}^{\infty} \mathsf{B}^n h \ \text{ for } \pi^T h = 0. \tag{18}$$

In matrix form, U is represented by the *fundamental matrix / group inverse* expression

$$\mathsf{U}h = (I - B + \Pi)^{-1} h, \qquad (\pi^T h = 0),$$

and one may record the deviation-matrix identity

$$(I - B)^{\#} := (I - B + \Pi)^{-1} - \Pi = \sum_{n=0}^{\infty} (B^n - \Pi),$$

where $(I - B)^{\#}$ is the group inverse of $I - B$ [6], and the series representation is the deviation-matrix identity [5].

Generalized microvalues. Given any integrable additive functional $Y_{k+1} = Y_k + \xi_{k+1}$ with conditional drift $\bar{\xi}(x) = \mathbb{E}[\xi_{k+1} \mid X_k = x]$ and stationary mean $\mu = \pi^T \bar{\xi}$, define the centered drift $h = \bar{\xi} - \mu\mathbf{1}$. Then Theorem 4.1 says that the unique martingale corrector is exactly the potential

$$g = \mathsf{U}h,$$

and the compensated process $Y_k - k\mu + g(X_k) - g(X_0)$ is a martingale. In this sense, the map $h \mapsto \mathsf{U}h$ is the canonical "micro-" transform associated with the Markov skeleton B: it takes a predictable drift profile and returns the unique centered correction that removes it.

The connection between the infinite-horizon corrector and finite-horizon partial sums is made precise by the following identity.

Finite-horizon identity. If h is centered and $g = \mathsf{U}h$ solves the Poisson equation $(I - B)g = h$ with $\pi^T g = 0$, then for every $n \geq 1$,

$$\sum_{\ell=0}^{n-1} B^\ell h \;=\; g - B^n g. \tag{19}$$

Since $B^n g \to \Pi g = 0$ as $n \to \infty$, (19) makes explicit that $g(X_k)$ is the horizon-free limit of the expected *centered* cumulative drift from time k onward.

4.4 Vector-valued functionals

All results extend componentwise to $\mathbb{R}^d$-valued rewards: one solves the same Poisson equation with multiple right-hand sides to obtain a vector martingale corrector.

5 Empirical highlights beyond the baseline

The classical microprice benchmark is well known, so the empirical question here is different: do the *new* constructions matter on the same data Stoikov used? We therefore reran the analysis on the public BAC and CVX March 2011 quote files released with the original notebook. We keep Stoikov's discretization choices (BAC: 10 imbalance bins and 2 spread buckets; CVX: 4 imbalance bins and 4 spread buckets) and his horizons (BAC: 1 and 3 minutes; CVX: 10 seconds).

5.1 Reproducing Stoikov on the public data

Figure 1 shows the BAC grouped conditional curves at Stoikov's 1- and 3-minute horizons. The familiar monotone relation between imbalance and future price change is present in both panels, and the longer horizon is easier to fit: the grouped RMSE for G^* falls from 8.92×10^{-4} at 1 minute to 5.47×10^{-4} at 3 minutes. On this centered BAC sample the weighted mid remains competitive, but the discounted resolvent already sits directly on top of the undiscounted correction.

Figure 2 shows the small-tick CVX case. Here the Poisson correction is the right state-based predictor: at 10 seconds the grouped RMSE is 6.97×10^{-4} for G^*, versus 9.10×10^{-4} for the one-step drift G_1, 1.83×10^{-3} for the raw mid, and 3.94×10^{-3} for the weighted mid. The figure also makes the centered-limit point concrete: Disc(.99) is again almost indistinguishable from G^* in every spread bucket.

These figures reproduce Stoikov's baseline on his own public data, but they also show the limit of what those files can answer. BAC and CVX are already close to the centered regime in which the undiscounted correction is expected to work, so they validate the classical construction more than they stress it.

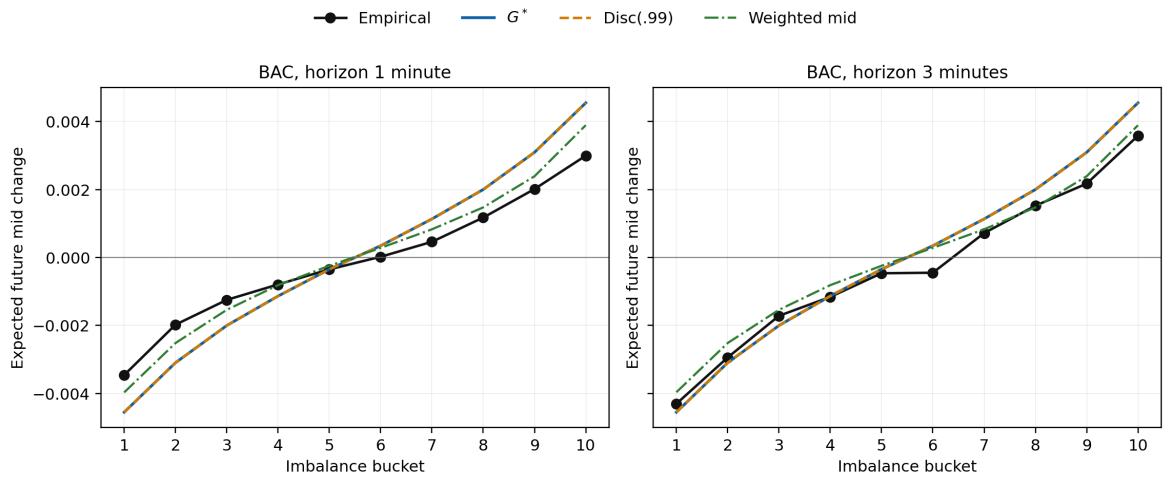

Figure 1: BAC March 2011 public data, using Stoikov's original discretization. The empirical conditional drift is monotone in imbalance, the fit improves from 1 to 3 minutes, and Disc(.99) is visually indistinguishable from G^*.

5.2 What is actually new

The replication figures anchor the paper, but they do not yet test the broader claim. For that we turn to the public WSELOB-2017 data: five large Warsaw Stock Exchange names, raw Level-I order-book events, and the same operator family estimated on the first 20 trading days of each file using a simple (spread, imbalance) state with 10 imbalance bins and 4 spread buckets. Figure 3 summarizes the out-of-sample grouped RMSE at 10 and 60 seconds together with the move-time martingale check.

Three points matter. First, the weighted mid is not a robust state-based value proxy outside Stoikov's original examples: on the Warsaw panel its grouped RMSE is 4.7× to 7.4× larger than G^* at 10 seconds, and still 2.1× to 6.6× larger at 60 seconds. Second, the finite-horizon decomposition predicts a genuine short-horizon crossover. At 10 seconds the raw mid is still the best grouped predictor on all five names, but by 60 seconds the Poisson correction overtakes it on every name, with grouped-RMSE improvements ranging from 25% to 126%. Third, the resolvent picture is not just formal: across all five Warsaw names the discounted operator sits on top of the centered one, with maximal grouped-RMSE gaps of only 1.5×10^{-5} at 10 seconds and 2×10^{-6} at 60 seconds.

The move-time statement is equally important. On the Warsaw panel the corrected price process remains close to a martingale on every name, with absolute move-time t-statistics between 0.68 and 1.97. This is the empirical counterpart of Theorem 4.1: the point is not merely that microprice "works," but that Stoikov's construction sits inside a larger centered-corrector family whose operator predictions survive out of sample on raw book data.

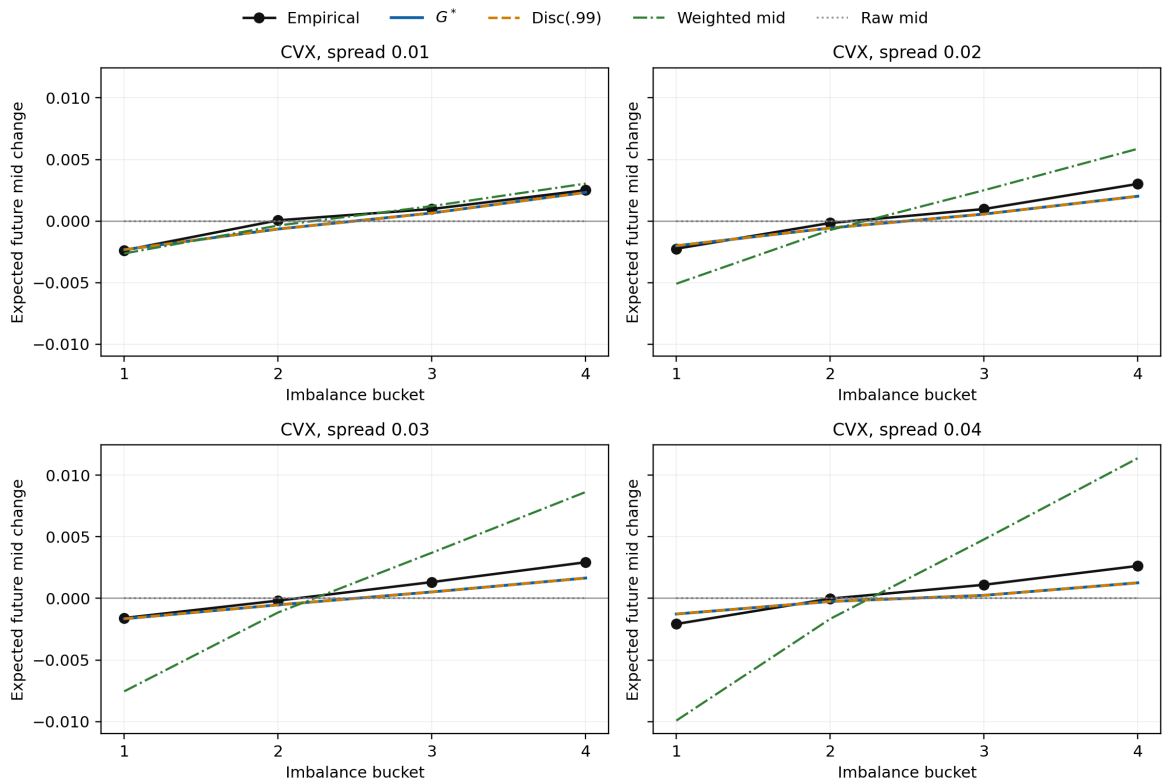

Figure 2: CVX March 2011 public data at the 10-second horizon. The post-move Poisson correction tracks the empirical conditional drift across spread buckets far better than the weighted mid, while the discounted resolvent remains on the same centered curve as G^*.

5.3 Venue-level stress check: public IEX TOPS

To check that the story is not an artifact of the two earlier datasets, we also decoded a full public IEX HIST TOPS PCAP for 2017-03-01 and reran the same coarse (spread, imbalance) pipeline on three liquid symbols (BAC, SPY, GLD). This is a deliberately harder setting: unlike Stoikov's notebook files and WSELOB, the feed is venue-local and only sees the top of the IEX book.

The result is informative precisely because it is mixed. BAC still behaves like a near-centered case, with move-time price $|t| = 0.16$ and a modest 60-second grouped-RMSE gain for G^* over both the raw mid and the weighted mid (2.49×10^{-2} versus 2.64×10^{-2} and 2.71×10^{-2}). SPY and GLD, however, fail the same centering diagnostic on this one-day venue-specific feed ($|t| = 3.09$ and 2.24), and the discounted resolvent is no longer a harmless boundary approximation. We therefore treat the IEX run as a stress test rather than as headline validation: it reinforces the paper's point that the centered-corrector picture is structurally right, but empirically fragile when the observed book is too local or the state is too coarse.

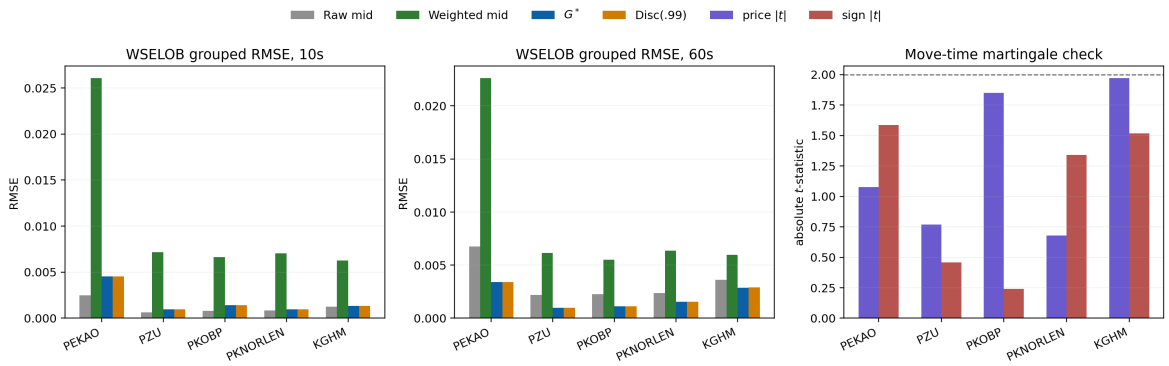

Figure 3: WSELOB-2017, five-name public raw-order-book panel (PEKAO, PZU, PKOBP, PKNORLEN, KGHM), using the first 20 trading days of each 2017 file. At 10 seconds the raw mid remains competitive, but the weighted mid is already far off the conditional drift. By 60 seconds G^* dominates both the weighted mid and the raw mid on every name, while Disc(.99) remains numerically indistinguishable from G^*. The price-corrected move-time process stays within the usual $|t| < 2$ band across all five names.

6 Discussion and limitations

The literature often uses "microprice" both for the instantaneous size-weighted mid m_t and for the long-horizon fair value $p_k + G^*(X_k)$; compare the discussion in [2]. Keeping these distinct avoids conceptual confusion.

The condition $\pi^T G_1 = 0$ asserts zero stationary drift per move and is substantive. If the drift is nonzero (trends, regime shifts), the undiscounted microprice is ill-posed; one should work with centered Poisson equations (removing the stationary drift as in Theorem 4.1) or with discounted resolvents.

Bounds based on $\|N\|_\infty \leq 1/\varepsilon$ require a uniform lower bound on move probabilities. In practice, some states may exhibit extremely rare moves, making N ill-conditioned and amplifying estimation error.

The framework is operator-theoretic: it applies to whatever finite-state Markov approximation is used. If the chosen state is not approximately Markov, the resulting kernels are unstable and predictive performance degrades; this is an empirical modelling issue rather than a limitation of the algebra.

7 Conclusion

Stoikov's microprice can be read as an operator identity rather than a collection of model variants. At mid-price move times, the mid-price is a Markov additive process; the microprice adjustment is the Poisson corrector on the post-move chain. Adding this corrector to the price produces a martingale, and the same Poisson equation constructs martingales for general additive order-book functionals.

The companion note develops the value-function extensions: discounted resolvents, Neumann-series representations, and the centered-limit interpretation of the microprice operator as the discount vanishes.

The public BAC/CVX rerun provides the anchor, but the Warsaw raw-book panel shows why the broader family matters. Stoikov's correction is one centered Poisson solution inside a larger operator class; on five additional public names the weighted mid is uniformly dominated, the raw mid only wins at the very shortest horizons, the Poisson correction takes over by 60 seconds, and the discounted resolvent collapses back onto the same centered object. The one-day IEX venue-level check then shows the boundary of that story: once the feed is local and centering weakens, the same coarse state can fail. The value-function viewpoint is therefore not a replacement claim; it is the stable continuation of Stoikov's operator once the centering condition weakens.

Several directions remain open: principled state-space selection, dynamic re-centering when $\pi^T G_1 \neq 0$, and continuous-time generator formulations that connect the discrete Poisson picture directly to CTMC microstructure models.

A Proofs

Proof of Lemma 3.2. For any matrix A, the spectral radius satisfies $\rho(A) \le \|A\|$ in any submultiplicative operator norm. Taking the induced ℓ^∞ norm,

$$\|Q\|_\infty = \max_x \sum_y |Q_{xy}| = \max_x \sum_y Q_{xy} \le 1 - \varepsilon.$$

Hence $\rho(Q) \le 1 - \varepsilon < 1$. □

Proof of Proposition 3.3. Since $\rho(Q) < 1$, the Neumann series $\sum_{k=0}^\infty Q^k$ converges absolutely and equals $(I - Q)^{-1}$. For the norm bound,

$$\|N\|_\infty \le \sum_{k=0}^\infty \|Q^k\|_\infty \le \sum_{k=0}^\infty \|Q\|_\infty^k \le \sum_{k=0}^\infty (1-\varepsilon)^k = \frac{1}{\varepsilon}.$$

The probabilistic interpretation is standard for absorbing chains: N_{xy} is the expected number of visits to y before absorption when starting from x; see, e.g., [5]. □

Proof of Lemma 3.5. From $N = I + Q + Q^2 + \cdots$ and non-negativity of Q, each term is non-negative entrywise. Thus $N \ge 0$ and $N_{xx} \ge 1$ for all x. □

Proof of Lemma 3.7. Let τ be the first event-time index at which a mid-price move occurs, as defined in Lemma 3.7. Conditioning on the first step from x gives the first-step equation

$$G_1(x) = \sum_y Q_{xy} G_1(y) + \sum_y T_{xy}\, \mathbb{E}[\Delta p \mid x \to y,\ \text{move}].$$

By definition, the second term is $\bar{r}_x$, hence $G_1 = \bar{r} + QG_1$ and $(I - Q)G_1 = \bar{r}$. Since $\rho(Q) < 1$, $I - Q$ is invertible and $G_1 = (I - Q)^{-1}\bar{r}$.

For the embedded chain, note that

$$\mathbb{P}_x(X_\tau = y) = \sum_{k=0}^\infty \mathbb{P}_x(X_k = y,\ \tau = k+1) = \sum_{k=0}^\infty (Q^k T)_{xy} = \big((I-Q)^{-1}T\big)_{xy} = B_{xy}.$$

□

Proof of Lemma 3.8. The move times (τ_k) are stopping times for the event-time filtration $(\mathcal{E}_n)$, and the event-time chain (X_n) is Markov with kernel $P = Q + T$. By the strong Markov property, the post-τ_k evolution of (X_n) depends on the past only through X_{τ_k}. The transition probabilities of the sampled chain follow:

$$\mathbb{P}(X_{\tau_{k+1}} = y \mid X_{\tau_k} = x) = \sum_{n=1}^\infty \mathbb{P}(X_{\tau_k+n} = y,\ \tau_{k+1} = \tau_k + n \mid X_{\tau_k} = x) = \sum_{n=0}^\infty (Q^n T)_{xy} = B_{xy},$$

where the last equality is the calculation from Lemma 3.7. □

Proof of Lemma 3.9. From $(Q+T)\mathbf{1} = \mathbf{1}$, we have $T\mathbf{1} = (I-Q)\mathbf{1}$. Thus

$$B\mathbf{1} = (I-Q)^{-1}T\mathbf{1} = (I-Q)^{-1}(I-Q)\mathbf{1} = \mathbf{1}.$$

Non-negativity of B follows from non-negativity of N and T. □

Proof of Proposition 3.17. By Lemma 3.7, $G_1(x) = \mathbb{E}_x[\Delta p_\tau]$. Assumption 2 implies $|\Delta p_\tau| \le D$ almost surely, so Jensen's inequality gives $|G_1(x)| \le \mathbb{E}_x[|\Delta p_\tau|] \le D$. Hence $\|G_1\|_\infty \le D$. For the operator-norm bound, $\|G_1\|_\infty = \|N\bar{r}\|_\infty \le \|N\|_\infty \|\bar{r}\|_\infty \le \|\bar{r}\|_\infty/\varepsilon$. □

Proof of Theorem 3.13. Write

$$B = \mathbf{1}\pi^T + C, \qquad C := B - \mathbf{1}\pi^T.$$

Since $B\mathbf{1} = \mathbf{1}$ and $\pi^T B = \pi^T$, we have $C\mathbf{1} = 0$ and $\pi^T C = 0$. Ergodicity implies that 1 is a simple eigenvalue of B and all other eigenvalues have modulus < 1, so $\rho(C) = \lambda_* < 1$.

Centering gives $\mathbf{1}\pi^T G_1 = 0$, hence for all $k \ge 0$,

$$B^k G_1 = (\mathbf{1}\pi^T + C)^k G_1 = C^k G_1.$$

Therefore,

$$\sum_{k=0}^{\infty} B^k G_1 = \sum_{k=0}^{\infty} C^k G_1 = (I-C)^{-1} G_1,$$

which proves the rank-one inverse formula since $I - C = I - B + \mathbf{1}\pi^T$. For centering, note that $\pi^T(I - B + \mathbf{1}\pi^T) = \pi^T$, so $\pi^T(I - B + \mathbf{1}\pi^T)^{-1} = \pi^T$, and therefore $\pi^T G^* = \pi^T G_1 = 0$.

To verify the Poisson equation, start from the partial sums $S_n := \sum_{k=0}^{n-1} B^k G_1$. Then

$$(I-B)S_n = G_1 - B^n G_1.$$

Since $B^n G_1 = C^n G_1 \to 0$ as $n \to \infty$ (because $\rho(C) < 1$), taking $n \to \infty$ yields $(I-B)G^* = G_1$. Uniqueness under the centering condition follows because the nullspace of $I - B$ is span$\{\mathbf{1}\}$: if $(I-B)g = 0$ and $\pi^T g = 0$, then g must be constant and hence $g \equiv 0$.

For truncation, note that $G^* - S_n = \sum_{k=n}^{\infty} C^k G_1$. In finite dimensions, for any $\rho \in (\rho(C), 1)$ there exists $C_\rho < \infty$ such that $\|C^k\|_\infty \le C_\rho \rho^k$ for all k. Hence

$$\|G^* - S_n\|_\infty \le \sum_{k=n}^{\infty} \|C^k\|_\infty \|G_1\|_\infty \le \sum_{k=n}^{\infty} C_\rho \rho^k \|G_1\|_\infty = \frac{C_\rho \rho^n}{1-\rho}\|G_1\|_\infty.$$

□

Proof of Proposition 3.14. Since $G^{(n)} = \sum_{\ell=0}^{n-1} B^\ell G_1$ and $G_1 = \mu\mathbf{1} + h$ with $\mu = \pi^T G_1$, we have

$$G^{(n)} = \sum_{\ell=0}^{n-1} B^\ell(\mu\mathbf{1} + h) = n\mu\,\mathbf{1} + \sum_{\ell=0}^{n-1} B^\ell h,$$

because $B^\ell \mathbf{1} = \mathbf{1}$ for every ℓ. Now $h = (I - B)\tilde{G} = \tilde{G} - B\tilde{G}$, so

$$\sum_{\ell=0}^{n-1} B^\ell h = \sum_{\ell=0}^{n-1} (B^\ell \tilde{G} - B^{\ell+1}\tilde{G}) = \tilde{G} - B^n \tilde{G}.$$

This yields (8). In the centered case $\mu = 0$ and the unique centered Poisson solution is $\tilde{G} = G^*$, giving the final identity. □

Proof of Theorem 3.15. Using the Markov property at move times,

$$\begin{aligned}\mathbb{E}[\mathcal{P}_{k+1} \mid \mathcal{F}_k] &= \mathbb{E}[p_{k+1} + G^*(X_{k+1}) \mid \mathcal{F}_k] \\ &= p_k + \mathbb{E}[\Delta p_{k+1} \mid X_k] + \mathbb{E}[G^*(X_{k+1}) \mid X_k] \\ &= p_k + G_1(X_k) + (BG^*)(X_k).\end{aligned}$$

From $(I - B)G^* = G_1$ we have $G_1 + BG^* = G^*$ pointwise, hence $\mathbb{E}[\mathcal{P}_{k+1} \mid \mathcal{F}_k] = p_k + G^*(X_k) = \mathcal{P}_k$. □

Proof of Theorem 4.1. Let $\mathcal{M}_k$ be defined by (17). Then

$$\mathcal{M}_{k+1} - \mathcal{M}_k = \xi_{k+1} - \mu + g(X_{k+1}) - g(X_k).$$

Taking conditional expectation given $\mathcal{G}_k$ and using $\mathbb{E}[\xi_{k+1} \mid \mathcal{G}_k] = \bar{\xi}(X_k)$ and $\mathbb{E}[g(X_{k+1}) \mid \mathcal{G}_k] = Bg(X_k)$ gives

$$\mathbb{E}[\mathcal{M}_{k+1} - \mathcal{M}_k \mid \mathcal{G}_k] = (\bar{\xi} - \mu\mathbf{1})(X_k) + (Bg - g)(X_k) = h(X_k) - (g - Bg)(X_k) = 0,$$

where the last equality is (16) written as $g - Bg = h$. Hence $(\mathcal{M}_k)$ is a martingale. □

Proof of Proposition 3.18. Write $Z = \hat{Z} + (Z - \hat{Z})$ and $Y = \hat{Z} + (Y - \hat{Z})$. Expanding,

$$(Z - Y)^2 = (Z - \hat{Z})^2 + (\hat{Z} - Y)^2 + 2(Z - \hat{Z})(\hat{Z} - Y).$$

Taking expectations, the cross term vanishes because $Y - \hat{Z}$ is $\mathcal{G}$-measurable and $Z - \hat{Z}$ is orthogonal to every $\mathcal{G}$-measurable square-integrable random variable:

$$\mathbb{E}\big[(Z - \hat{Z})(Y - \hat{Z})\big] = \mathbb{E}\big[\mathbb{E}[Z - \hat{Z} \mid \mathcal{G}]\,(Y - \hat{Z})\big] = 0.$$

This yields (10). □

Proof of Corollary 3.19. Apply Proposition 3.18 with $\mathcal{G} = \mathcal{F}_k$, $Z = p_{k+n}$, $\hat{Z} = p_k^{(n)} = \mathbb{E}[p_{k+n} \mid \mathcal{F}_k]$, and $Y = \hat{p}_k = p_k + a(X_k)$. Then (10) becomes

$$\mathbb{E}[(p_{k+n} - \hat{p}_k)^2] = \mathbb{E}[(p_{k+n} - p_k^{(n)})^2] + \mathbb{E}[(p_k^{(n)} - \hat{p}_k)^2].$$

Using the decomposition $p_k^{(n)} = p_k + G^{(n)}(X_k)$ from (5) gives $p_k^{(n)} - \hat{p}_k = G^{(n)}(X_k) - a(X_k)$, yielding (11). The optimality statements follow immediately because the second term is a nonnegative quadratic functional of a. □

Proof of Proposition 3.22. For (13), expand the discounted sum and use the Markov property at move times:

$$\mathbb{E}[\Delta p_{k+j} \mid \mathcal{F}_k] = \mathbb{E}\big[\mathbb{E}[\Delta p_{k+j} \mid X_{k+j-1}] \,\big|\, \mathcal{F}_k\big] = \mathbb{E}\big[G_1(X_{k+j-1}) \,\big|\, \mathcal{F}_k\big] = (B^{j-1}G_1)(X_k).$$

Therefore

$$\mathbb{E}\left[\sum_{j=1}^{\infty} \alpha^{j-1}\Delta p_{k+j} \,\middle|\, \mathcal{F}_k\right] = \sum_{j=1}^{\infty} \alpha^{j-1}(B^{j-1}G_1)(X_k) = \sum_{\ell=0}^{\infty} \alpha^{\ell}(B^{\ell}G_1)(X_k) = G^{(\alpha)}(X_k).$$

For the geometric-horizon identity (14), use the tower property and the decomposition $p_{k+n} = p_k + \sum_{j=1}^{n} \Delta p_{k+j}$:

$$\begin{aligned}\mathbb{E}[p_{k+N} \mid \mathcal{F}_k] &= \sum_{n=0}^{\infty}(1-\alpha)\alpha^n\, \mathbb{E}[p_{k+n} \mid \mathcal{F}_k] \\ &= p_k + \sum_{n=1}^{\infty}(1-\alpha)\alpha^n \sum_{j=1}^{n} \mathbb{E}[\Delta p_{k+j} \mid \mathcal{F}_k].\end{aligned}$$

Swap the order of summation (Fubini is justified since the increments are bounded) to obtain

$$\sum_{n=1}^{\infty}(1-\alpha)\alpha^n \sum_{j=1}^{n} \mathbb{E}[\Delta p_{k+j} \mid \mathcal{F}_k] = \sum_{j=1}^{\infty}\left(\sum_{n=j}^{\infty}(1-\alpha)\alpha^n\right)\mathbb{E}[\Delta p_{k+j} \mid \mathcal{F}_k] = \sum_{j=1}^{\infty} \alpha^j\, \mathbb{E}[\Delta p_{k+j} \mid \mathcal{F}_k].$$

Finally, apply (13) and factor out α to get (14). □

Proof of Proposition 3.23. Since $\pi^T G_1 = 0$, we have $B^k G_1 = C^k G_1$ with $C := B - \mathbf{1}\pi^T$ and hence $\|B^k G_1\|_\infty \le C_\rho \rho^k \|G_1\|_\infty$ for all k. Using the series definitions,

$$G^* - G^{(\alpha)} = \sum_{k=0}^{\infty}(1-\alpha^k)B^k G_1.$$

Because $1 - \alpha^k \le k(1-\alpha)$ for $\alpha \in (0,1)$,

$$\|G^* - G^{(\alpha)}\|_\infty \le \sum_{k=0}^{\infty}(1-\alpha^k)\,\|B^k G_1\|_\infty \le (1-\alpha)\sum_{k=0}^{\infty} k\, C_\rho \rho^k\, \|G_1\|_\infty.$$

The power-series identity $\sum_{k=0}^{\infty} k\rho^k = \rho/(1-\rho)^2$ (for $|\rho| < 1$) yields the claimed bound. □

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